

How Much Does the Activation Function Matter? A 2,000-Run Variance Study of Gated Recurrent Networks for Sentiment Classification

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Abstract—Activation-function comparisons in gated recurrent networks are almost always reported from a single training run. It has therefore never been established whether the differences they report are larger than the noise of retraining the same model. This paper settles that question at scale. We train all 100 ordered pairs of ten activation functions in the dense layers of four gated recurrent architectures on IMDb, and repeat every one of the 400 configurations under five random seeds, for 2,000 models in total with every other setting held fixed. A variance decomposition over these runs shows that the random seed accounts for 58.7% of all variation in test accuracy and the architecture for 26.7%, while the activation pair explains only 3.7%. The consequence is direct: single-seed activation rankings do not reproduce. Across the four architectures the Spearman correlation between the single-seed ranking of the 100 pairs and their five-seed means is -0.174 , -0.059 , -0.041 and $+0.055$, indistinguishable from zero in every case, and the ten best configurations overlap within their confidence intervals. What survives is architectural: GRU outperforms LSTM by 0.79 points ($p < 10^{-100}$, Cohen’s $d = 1.19$), supplies all fifty highest-scoring configurations, and is the more reliable choice, with roughly 40% lower seed-to-seed standard deviation. One activation effect also survives: smooth functions reach 73.7% validation accuracy after one epoch against 61.1% for the ReLU family. We give the seed budget needed to resolve each remaining effect, and argue that grid searches of this kind should report variance as standard.

Index Terms—Sentiment Analysis, IMDb Dataset, Deep Learning, LSTM, GRU, Activation Functions, Reproducibility, Seed Variance, Experimental Methodology.

I. INTRODUCTION

Sentiment analysis is the task of deciding whether a piece of text is positive or negative. It is one of the most common problems in natural language processing. The IMDb Large Movie Review Dataset, with 50,000 balanced reviews, is the standard benchmark for the binary version of the task [16]. LSTM and GRU networks are the usual choice for this problem because their gating mechanisms let them track context across long sequences [1]–[3].

One part of these models gets little attention: the activation function in the dense layers that sit after the recurrent layers.

The gates inside the recurrent cell are fixed to sigmoid and tanh, but the dense layers that turn the recurrent output into a final prediction can use any activation function. Ahmed et al. [17] tested three classical combinations in a bidirectional LSTM, reached 88.40% accuracy, and asked for follow-up work on modern activation functions and other architectures.

Answering that call runs into a problem. Every comparison we are aware of, including all of [17], [18], [21]–[23], trains each configuration once. Training is not deterministic. The weight initialisation, the dropout masks and the order of the training batches all change the final accuracy, even when the data and every setting are held identical. If that change is as large as the gaps between activation functions, then a ranking built from one run per function is measuring the seed rather than the function. No study in this area reports how large that change is, so there is no way to tell which is being measured.

We therefore designed this study around replication. We train all 100 ordered pairs of ten activation functions across four architectures (LSTM, GRU, and the fully bidirectional version of each), then repeat every one of those 400 configurations under five random seeds, holding the data split fixed so that only initialisation, dropout and batch order vary. The result is **2,000 trained models**. To our knowledge this is the largest controlled activation study in gated recurrent networks, and the first in this setting to report seed variance. Five runs of every configuration let us measure how much accuracy moves when only the seed changes, and compare that against the gaps between activation functions:

- **RQ1:** How is accuracy variance divided between architecture, activation choice and seed?
- **RQ2:** Does the architecture exceed seed noise?
- **RQ3:** Does the activation pair exceed seed noise?
- **RQ4:** Does a single-seed ranking reproduce?
- **RQ5:** Which activation effects survive?
- **RQ6:** How many seeds would the rest require?

II. LITERATURE REVIEW

Hochreiter and Schmidhuber [1] introduced the LSTM to address the vanishing gradient problem in recurrent networks,

using sigmoid and tanh within the recurrent cell. Cho et al. [2] proposed the GRU, which uses two gates and combines the cell and hidden states. Chung et al. [3] compared GRU and LSTM on sequence-modeling tasks and found GRU competitive with LSTM. Schuster and Paliwal [4] introduced bidirectional recurrent networks, allowing models to use both past and future context. These studies established the recurrent architectures used in our comparison.

The activation function literature developed alongside these architectural advances. Nair and Hinton [5] studied ReLU, which avoids saturation for positive inputs but has zero gradients for negative inputs. Maas et al. [6] and Clevert et al. [8] studied LeakyReLU and ELU, which allow nonzero outputs for negative inputs. Klambauer et al. [9] introduced SELU with self-normalising properties under specific conditions. Later work introduced GELU [10], Swish [11], and Mish [12], while surveys [14], [15] reviewed the growing range of activation functions. These developments motivate examining their effects in the dense layers following gated recurrent networks.

Within sentiment analysis, IMDb [16] is a widely used binary classification benchmark. Tang et al. [20] studied gated recurrent networks for document-level sentiment classification, and Socher et al. [19] introduced the Stanford Sentiment Treebank. Several studies have examined activation choice. Eger et al. [21] compared 21 activations across eight NLP tasks and also investigated alternative activations within LSTM cells. Farzad et al. [22] compared 23 activation functions in a basic LSTM network on IMDb, Movie Review, and MNIST. Essai Ali et al. [23] evaluated 26 state activation functions in LSTM networks on classification tasks outside sentiment analysis. Zhu and Tan [18] tested a custom activation in simple RNN, LSTM, and GRU models on IMDb, comparing it with ReLU, LeakyReLU, ELU, SELU, and tanh. Ahmed et al. [17] compared three classical activation pairings in a bidirectional LSTM, reported 88.40% accuracy, and suggested examining modern functions and other architectures.

Our study brings these comparisons into a shared experimental design. We evaluate all 100 ordered pairs of ten activation functions in two dense layers across four architectures. These use LSTM or GRU cells, a bidirectional first recurrent layer, and either a unidirectional or bidirectional second layer. The comparison includes GELU, SiLU, and Mish, with the data split and training settings held fixed.

Repeated training is also part of earlier work. Eger et al. [21] explicitly averaged results over five random weight initializations in their sentence-classification experiments. Our main contribution is therefore the analysis of how variation between training runs compares with the effects of architecture and dense-layer activation choice in this setting. We train every configuration with five seeds and examine whether rankings from one run agree with rankings based on repeated runs. This allows us to assess which performance differences are clear and which remain uncertain.

III. METHODOLOGY

A. Dataset and Preprocessing

The IMDb Large Movie Review Dataset [16] (50,000 reviews, balanced binary labels) was split 80/20 train-test with `random_state=42`, and 10% of training data was withheld as a validation set. Reviews were cleaned by removing HTML tags and non-alphabetic characters, then lowercased. A vocabulary of the 5,000 most frequent training-set tokens was used; all sequences were truncated or zero-padded to 200 tokens.

B. Network Architecture

All four experiments share the same model structure. The only thing that changes within each experiment is the choice of Act_1 and Act_2 in the two dense layers. The shared architecture is: an embedding layer (vocabulary 5,000, dimension 128), a bidirectional recurrent Layer 1 (64 units per direction), a recurrent Layer 2 (32 units per direction), Dropout(0.4), Dense Layer 1 (64 units) with Act_1 , Dense Layer 2 (32 units) with Act_2 , and a single output unit trained with BCEWithLogitsLoss. The four experiments differ only in the recurrent cell type (LSTM, GRU) and whether Layer 2 is bidirectional (Table I).

TABLE I
EXPERIMENTAL ARCHITECTURE SUMMARY

Exp.	Layer 1	Layer 2	Params
1	BiLSTM(64)	LSTM(32)	$\approx 764\text{K}$
2	BiGRU(64)	GRU(32)	$\approx 734\text{K}$
3	BiLSTM(64)	BiLSTM(32)	$\approx 787\text{K}$
4	BiGRU(64)	BiGRU(32)	$\approx 752\text{K}$

C. Activation Functions Under Investigation

Ten activation functions were evaluated (Table II). The classical family is ReLU [5], LeakyReLU [6], PReLU [7], ELU [8], Tanh and SELU [9]. The modern smooth family is GELU [10], SiLU (Swish) [11], Mish [12] and Hardswish [13]. The latter four had not previously been benchmarked in gated recurrent dense layers for sentiment classification.

TABLE II
ACTIVATION FUNCTIONS: FAMILIES AND KEY PROPERTIES

Function	Type	Formula / Key Property
ReLU	Classical	$\max(0, x)$; dead neurons for $x < 0$
ELU	Classical	$\alpha(e^x - 1)$ for $x \leq 0$; mean-centering
LeakyReLU	Classical	$0.01x$ for $x < 0$; low dead-neuron risk
Tanh	Classical	Bounded; saturates for $ x \gg 1$
GELU	Modern	$x \cdot \Phi(x)$; probabilistic gating
SiLU	Modern	$x \cdot \sigma(x)$; self-gating
Mish	Modern	$x \cdot \tanh(\text{softplus}(x))$; smooth
SELU	Modern	Self-normalising; implicit batch norm
PReLU	Modern	Learnable slope α ; input-dependent
Hardswish	Modern	Piecewise; dead zone for $x < -3$

D. Training Configuration and Evaluation

Every setting in Table III was held the same across all runs, so the activation pair is the only thing that changes between models. The architecture parameters (vocabulary, embedding, layer sizes, dropout) replicate Ahmed et al. [17] exactly. The training parameters (optimizer, learning rate, batch size, scheduler, clipping) were chosen by us because Ahmed et al. did not report them. After training, accuracy, precision, recall, and F1-score were measured on the held-out 10,000-review test set.

E. Replication Protocol

Each of the $10 \times 10 \times 4 = 400$ configurations was trained under five seeds, $\{10, 20, 30, 40, 50\}$, giving 2,000 models. The seed is applied immediately before the model and its data loaders are constructed, setting `random`, `numpy`, `torch` and `torch.cuda` with `cudnn.deterministic=True`, so a given (pair, seed) is exactly reproducible.

The split is deliberately *not* a function of the seed: it stays pinned at `random_state=42`. The variance we report is therefore the variance of retraining the same model on the same data. That is the quantity a single-run comparison assumes to be negligible. Split variance would only inflate it, so these estimates are conservative.

Runs were executed five at a time, one worker per seed. Each worker is a separate process with its own RNG and deterministic kernel selection, so concurrency cannot perturb a result; we confirmed this by re-running sampled configurations serially and recovering identical metrics.

TABLE III
SHARED HYPERPARAMETER CONFIGURATION

Hyperparameter	Value	Source
Vocab / Seq. length / Embed.	5,000 / 200 / 128	[17]
RNN Layer 1 / Layer 2 units	64 / 32 per direction	[17]
Dense Layer 1 / Layer 2 units	64 / 32	[17]
Dropout rate	0.4	[17]
Batch size / Max epochs	64 / 10	Ours
Early stopping patience	3 epochs (val. loss)	Ours
Optimiser	Adam, $\text{lr} = 10^{-3}$	Ours
LR scheduler	ReduceLROnPlateau ($\times 0.5$)	Ours
Loss / Gradient clipping	BCEWithLogitsLoss / $L_2 \leq 1.0$	Ours
Weight initialisation	Kaiming uniform (Linear)	PyTorch default
Data split seed	42, fixed for all runs	Ours
Training seeds	10, 20, 30, 40, 50	Ours

F. On Reproducing the Baseline of Ahmed et al.

Ahmed et al. [17] describe their architecture in full and we copy it exactly, but they do not report how they trained it: optimizer, learning rate, batch size, epochs, early stopping, seed and data split are all missing. Exact reproduction is therefore impossible, and we adopt the fully documented settings of Table III instead.

Replication sharpens what can be said about their 88.40%. Our best configuration averages 87.81% with a 95% upper bound of 88.01%, and no run among the 2,000 reached 88.40% (maximum 88.39%), so the benchmark sits outside the interval

of our strongest setting. Since they report a single run without a seed or standard deviation, and we measure the per-cell standard deviation of this model family at 0.38 points, their figure is best read as one draw of unknown uncertainty. We treat it as a reference point, not a target.

IV. RESULTS

A. Where the Variance Comes From

Test accuracy differs from run to run for three reasons: the architecture, the activation pair, and the random seed. Table IV splits the total variance across the 2,000 runs between them.

The seed accounts for **58.7%**. This is variance between runs that differ in nothing but the seed, so no choice of architecture or activation removes it. The architecture accounts for 26.7%, almost all of it (26.3%) from the choice of LSTM or GRU cell; making the second recurrent layer bidirectional accounts for 0.19%. The activation pair accounts for **3.7%**.

TABLE IV
VARIANCE DECOMPOSITION OF TEST ACCURACY (2,000 RUNS)

Source of variation	% of total variance
Architecture (4 levels)	26.69
cell type (GRU vs. LSTM)	26.32
layer-2 bidirectionality	0.19
Act ₁ (10 levels)	0.53
Act ₂ (10 levels)	0.55
Act ₁ × Act ₂ pair (100 levels)	3.74
Seed (residual)	58.67

The activation effect is about sixteen times smaller than the seed effect. A study that trains each configuration once measures the two together and cannot separate them. Section V-C gives the number of seeds each effect needs.

B. Classification Performance

Fig. 1 plots the mean accuracy of each of the 400 configurations, grouped by architecture. Every architecture mean comes from 500 runs, which gives a 95% confidence interval of ± 0.03 to ± 0.10 points.

The two GRU architectures both average 87.48% and cannot be told apart. Both are above the two LSTM architectures, at 86.62% and 86.75%. Within each architecture the 100 activation pairs form one narrow cloud, and the gap between the GRU and LSTM clouds is wider than any cloud. Changing the cell type moves accuracy more than changing the activation pair does.

Pooling by cell type over 1,000 runs each, GRU reaches $87.48 \pm 0.02\%$ and LSTM $86.69 \pm 0.05\%$. The gap of **0.79** points is large relative to its uncertainty (Welch $t = 26.7$, $p \approx 2 \times 10^{-125}$, Cohen's $d = 1.19$). No configuration's mean reaches the 88.40% benchmark of Ahmed et al. [17]; the best cell mean is 87.81% with a 95% upper bound of 88.01%, and none of the 2,000 individual runs attained 88.40% (maximum 88.39%). We therefore treat that figure as unmatched under our documented training settings rather than as a target we reached.

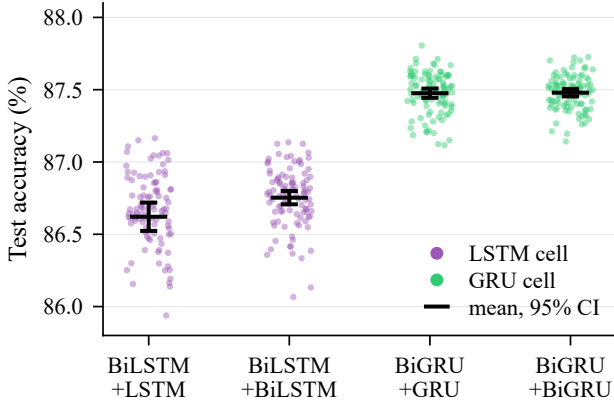


Fig. 1. The 400 replicated cell means, grouped by architecture. Each point is one activation pair averaged over five seeds; black markers give the architecture mean with its 95% confidence interval. The y -axis is truncated at 85.8%, which clips one outlier cell (Hardswish + LeakyReLU in BiLSTM+LSTM, 82.06%, $\sigma = 9.27$).

C. The Ten Best Configurations

Table V lists the ten highest-scoring configurations of the 400, each with its standard deviation and 95% confidence interval. All ten use a GRU cell, as do the top fifty.

The ten means lie between 87.68% and 87.81%, a spread of **0.12** points. Their confidence intervals are wider than that, up to ± 0.50 points. Every interval overlaps every other, and one range of values is consistent with all ten at once. *Nothing in the data separates these ten configurations*, so the order of the rows in Table V is not a ranking.

TABLE V
TEN HIGHEST-ACCURACY CONFIGURATIONS OF 400, WITH SEED VARIABILITY

Act ₁ + Act ₂	Experiment	Acc.% mean $\pm \sigma$	95% CI $\pm$	F1% mean
ReLU + LeakyReLU	BiGRU+GRU	87.81 $\pm$ 0.17	0.21	87.80
SiLU + Mish	BiGRU+BiGRU	87.73 $\pm$ 0.16	0.20	87.65
Tanh + LeakyReLU	BiGRU+BiGRU	87.72 $\pm$ 0.40	0.50	87.70
Tanh + GELU	BiGRU+GRU	87.71 $\pm$ 0.29	0.36	87.75
PReLU + ReLU	BiGRU+GRU	87.71 $\pm$ 0.17	0.22	87.67
GELU + GELU	BiGRU+BiGRU	87.70 $\pm$ 0.18	0.22	87.58
SiLU + Hardswish	BiGRU+BiGRU	87.70 $\pm$ 0.10	0.12	87.69
GELU + Hardswish	BiGRU+BiGRU	87.69 $\pm$ 0.18	0.23	87.65
SiLU + GELU	BiGRU+GRU	87.68 $\pm$ 0.18	0.22	87.63
PReLU + GELU	BiGRU+GRU	87.68 $\pm$ 0.28	0.34	87.71

All ten are GRU-based. The ten means span 0.12 points, less than the widest half-interval (± 0.50); every pair of intervals overlaps.

D. Reproducibility of the Single-Seed Ranking

We ranked the 100 activation pairs in each architecture twice: once by the accuracy of the original single run, and once by the five-seed mean. Table VI compares the two rankings.

The Spearman correlation between the two rankings is -0.174 , -0.059 , -0.041 and $+0.055$ in the four architectures. A correlation of zero means the first ranking says nothing about the second, and all four values are close to zero. A pair moves 32 to 37 places out of 100 on average between the two

TABLE VI
AGREEMENT BETWEEN THE SINGLE-SEED RANKING AND THE 5-SEED MEANS

Experiment	ρ	Mean rank Δ	Top-10 overlap	Mean $ \Delta\text{acc} $
BiLSTM+LSTM	-0.174	36.9	0/10	0.53
BiGRU+GRU	-0.059	34.5	1/10	0.34
BiLSTM+BiLSTM	-0.041	34.4	1/10	0.45
BiGRU+BiGRU	$+0.055$	32.1	0/10	0.36

ρ is the Spearman correlation between the 100 single-seed cell accuracies and the corresponding 5-seed means. $\rho \approx 0$ in all four.

lists. Of the ten pairs that led each single-seed sweep, at most one is still in the top ten after replication.

The cause is selection bias. A pair reaches the top of a single-seed sweep partly because it is good and partly because its one run was lucky. LeakyReLU + Tanh led the BiGRU+GRU sweep at 88.05%. Its five runs are 86.70, 87.13, 87.45, 87.46 and 88.20, so the published figure was its best draw; its mean is 87.39 ± 0.55 , which ranks 77th of 100. The pair that leads on the mean, ReLU + LeakyReLU, scores 87.81 ± 0.17 . The single-seed winners of the other three architectures fall to ranks 98, 91 and 85. Averaged over the ten leaders of the original sweep, replication costs 0.48 points.

E. Marginal Effects per Activation

Averaging each function over all ten of its partners and all five seeds gives 50 runs per estimate, which removes most of the noise. A small signal appears at that level: the ten averages within one architecture span at most 0.67 points, and within BiGRU+GRU only 0.18.

The orderings still do not agree between architectures. Comparing the four architectures pairwise gives six rank correlations, from -0.164 to $+0.576$, none significant at the 5% level ($p \geq 0.082$). A single-seed reading of this grid suggested that a function good in one architecture was bad in another, with ELU as the example: best Act₁ in BiGRU+GRU and last in BiLSTM+LSTM. Replicated, ELU ranks 9th of ten in BiGRU+GRU and 3rd in BiLSTM+LSTM, so both halves of that claim reverse. The orderings are unrelated rather than inverted. Per-function averages for all four architectures are given in the supplementary material.

F. Convergence Speed

The main sweep records only final accuracy. To measure learning speed we re-ran the ten homogeneous pairs (X+X) on BiGRU+GRU with per-epoch logging under five seeds (Table VII, Fig. 2).

This is the one activation effect in the study that survives replication. After a single epoch the smooth family (Mish, GELU, SiLU, Hardswish) averages **73.70%** validation accuracy against **61.05%** for the ReLU family (ReLU, LeakyReLU, SELU). The 12.65-point gap is 2.9 times the median per-function seed standard deviation of 4.39, and the ordering is preserved against the single-seed measurement (Spearman $\rho = 0.891$, $p = 0.0005$). The advantage is in speed only: by

TABLE VII
EPOCH-1 CONVERGENCE ON BiGRU+GRU, FIVE SEEDS

Activation	Val. loss (E1)	Val. acc. (E1)%	σ
Hardswish	0.523	74.55	4.05
GELU	0.541	74.21	3.32
Mish	0.544	74.03	5.97
SiLU	0.558	72.00	3.95
PReLU	0.625	66.87	2.22
Tanh	0.651	62.76	7.52
LeakyReLU	0.655	62.40	4.73
ELU	0.644	62.20	7.29
ReLU	0.652	61.51	7.50
SELU	0.674	59.24	3.73

Smooth family (Mish, GELU, SiLU, Hardswish) averages 73.70%; the ReLU family (ReLU, LeakyReLU, SELU) averages 61.05%.

epoch three all functions reach comparable validation loss, and the final test accuracies are within noise of each other.

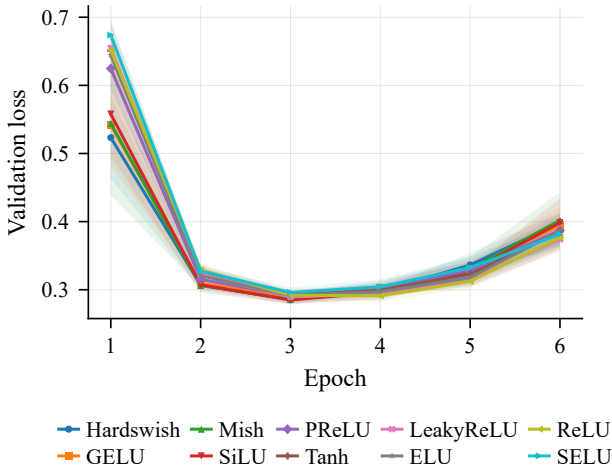


Fig. 2. Per-epoch validation loss on BiGRU+GRU for the ten homogeneous pairs. Curves are means over five seeds; shaded bands are ± 1 standard deviation. The separation visible at epoch 1 has closed by epoch 3.

G. Reliability and Failure Modes

Accuracy is not the only axis on which the architectures differ. The median per-cell standard deviation is 0.30 for the GRU family against 0.50 for the LSTM family ($p \approx 2 \times 10^{-7}$): GRU is both more accurate and about 40% more stable run-to-run. Instability is also concentrated. Of the eight cells whose standard deviation exceeds 1.0 point, *all eight* are LSTM-family. The extreme case, Hardswish + LeakyReLU in BiLSTM+LSTM, averages 82.06% with $\sigma = 9.27$ and a range of 65.50–86.85%: it trains normally under four seeds and collapses under the fifth. Apart from that collapse, no run of the 2,000 fell below 60% accuracy.

Two claims from the single-seed reading do not survive. Per-cell precision–recall imbalance is one: the widest gap between mean precision and mean recall over any cell is 6.89 points while individual runs reach 24.17, so large gaps belong to runs, not to configurations. A dense-layer ReLU penalty is the

other; in BiLSTM+LSTM, ReLU is the strongest Act₁ choice (86.83% versus 86.60%, $p = 0.006$).

The campaign cost about six GPU-hours on one NVIDIA RTX 5080 at 13.4 s per run. GRU models carry 3.8% fewer parameters than their LSTM counterparts (734K versus 764K), making BiGRU+GRU the cheapest, joint most accurate and most stable of the four.

V. DISCUSSION

A. RQ1–RQ3: Architecture, Activation Choice and Training Seeds

Training seeds account for 58.7% of the variation in test accuracy, compared with 26.7% for architecture and 3.7% for activation pairs. In this experiment, accuracy varies more between training runs than with either architecture or activation choice alone.

GRU averages 0.79 percentage points above LSTM, with Cohen’s $d = 1.19$. Architecture means change by at most 0.18 points between the single-seed and five-seed results. Making the second recurrent layer bidirectional accounts for 0.19% of the variance. It adds 0.13 points in LSTM ($p = 0.017$), while the measured change in GRU rounds to 0.00 points ($p = 0.858$). The GRU comparison therefore shows no clear accuracy gain from a bidirectional second layer.

Within BiGRU+GRU, the five-seed means of the 100 activation pairs span 0.69 percentage points. The median standard deviation across seeds is 0.33 points, showing that variation between runs is substantial compared with the differences between pairs. The full 0.69-point spread is also smaller than the 0.79-point gap between GRU and LSTM. These comparisons show a clearer difference between the two model families than between individual activation pairs.

B. RQ4: Agreement Between Single-Seed and Five-Seed Rankings

The two rankings show little agreement in all four architectures (Table VI). Their Spearman correlations range from -0.174 to $+0.055$, all close to zero. Activation pairs move 32 to 37 places on average between the rankings. In each architecture, at most one pair from the original top ten remains in the top ten after averaging five runs.

The small accuracy gaps between pairs help explain these changes. Across architectures, the median standard deviation across seeds ranges from 0.29 to 0.53 percentage points. When accuracy gaps are small relative to this variation, changes between training runs can change the order of the pairs. Here, the ranking from one run gives little indication of the ranking based on five runs.

C. RQ5–RQ6: Findings Across Seeds and Estimated Seed Counts

The clearest activation difference appears after the first epoch. Mish, GELU, SiLU and Hardswish average 73.70% validation accuracy, compared with 61.05% for ReLU, LeakyReLU and SELU. The 12.65-point gap is 2.9 times the median standard deviation across seeds. The single-seed and five-seed rankings

are also similar ($\rho = 0.891$). This difference concerns early learning; final accuracy shows no clear separation. GRU models vary less between runs than LSTM models. All eight configurations with a standard deviation above one percentage point use LSTM. The estimated seed counts depend on the size of the accuracy gap. Using the median standard deviation of 0.38 percentage points, 80% power and $\alpha = 0.05$, the 0.79-point gap between GRU and LSTM corresponds to about 4 seeds per configuration. The 0.12-point gap between the first- and tenth-ranked configurations corresponds to about 157. These estimates are based on the observed variation. At 13.4 s per run, 157 seeds across 100 activation pairs correspond to about 58 GPU-hours for one architecture.

D. Limitations

The study covers one dataset and one task family, so the variance split should be re-estimated before being assumed elsewhere. The data split is held fixed by design, so our estimates exclude split variance and are lower bounds. Five seeds suffice for the architectural conclusions, but leave per-cell intervals wide relative to the differences between cells, so individual cell means should be read with their intervals attached.

VI. CONCLUSION

We trained 2,000 models, testing 100 activation pairs across four recurrent architectures with five seeds each. Training seeds accounted for 58.7% of the variation in test accuracy, compared with 26.7% for architecture and 3.7% for activation pairs.

Our main contribution is showing how repeated training changes the conclusions of an activation comparison. Earlier studies discussed in this paper reported rankings from single runs. In our experiments, those rankings showed little agreement with five-run rankings, moving more than 32 places on average. This challenges the use of a single run as evidence that one activation pair is better than another.

Architecture differences were clearer. GRU exceeded LSTM by 0.79 percentage points, supplied all fifty highest-scoring configurations, and showed about 40% lower standard deviation across seeds. Making the second GRU layer bidirectional added no measurable accuracy benefit.

Activation choice mainly affected early learning. Mish, GELU, SiLU and Hardswish averaged 73.70% validation accuracy after one epoch, compared with 61.05% for ReLU, LeakyReLU and SELU. Their early advantage did not produce a clear difference in final test accuracy.

Under our IMDb settings, architecture choice deserves more attention than small differences between activation pairs. More broadly, claims of better performance should be supported by repeated runs and reported uncertainty. Future work should test these findings on other datasets, different data splits, and attention-based models.

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